



## Chapter 7

CS 60 (University of California Los Angeles)



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#### 56 CODE

Write a query that displays the book title, cost and year of publication for every book in the system. Sort the results by book title.

```
SELECT BOOK_TITLE, BOOK_COST, BOOK_YEAR  
FROM BOOK  
ORDER BY BOOK_TITLE;
```

#### 57 CODE

Write a query that displays the first and last name of every patron, sorted by last name and then first name. Ensure the sort is case insensitive (*Figure P7.57*). (50 rows)

```
SELECT PAT_FNAME, PAT_LNAME  
FROM PATRON  
ORDER BY PAT_LNAME, PAT_FNAME;
```

#### 58 CODE

Write a query to display the checkout number, checkout date, and due date for every book that has been checked out sorted by checkout number (*Figure P7.58*). (68 rows)

```
SELECT CHECK_NUM, CHECK_OUT_DATE, CHECK_DUE_DATE  
FROM CHECKOUT  
ORDER BY CHECK_NUM;
```

#### 59 CODE

Write a query to display the book number, book title, and subject for every book sorted by book number (*Figure P7.59*). (20 rows)

```
SELECT BOOK_NUM, BOOK_TITLE AS TITLE, BOOK_SUBJECT AS "Subject of Book"  
FROM BOOK  
ORDER BY BOOK_NUM;
```

#### 60 CODE

Write a query to display the different years in which books have been published. Include each year only once and sort the results by year (*Figure P7.60*).

```
SELECT DISTINCT BOOK_YEAR  
FROM BOOK  
ORDER BY BOOK_YEAR;
```

#### 61 CODE

Write a query to display the checkout number, book number, patron ID, checkout date, and due date for every checkout that has ever occurred in the system. Sort the results by checkout date in descending order and then by checkout number in ascending order (Partial results are shown in *Figure P7.63*).

```
SELECT CHECK_NUM, BOOK_NUM, PAT_ID, CHECK_OUT_DATE,  
CHECK_DUE_DATE  
FROM CHECKOUT  
ORDER BY CHECK_OUT_DATE DESC, CHECK_NUM;
```

#### 62 CODE

Write a query to display the book number, title, and cost of each book sorted by book number (Partial results are shown in *Figure P7.62*).

```
SELECT BOOK_NUM, BOOK_TITLE, BOOK_COST  
FROM BOOK  
ORDER BY BOOK_NUM;
```

#### CODE 63

Write a query to display the checkout number, book number, patron ID, checkout date, and due date for every checkout that has ever occurred in the system. Sort the results by checkout date in descending order and then by checkout number in ascending order (Partial results are shown in *Figure P7.63*)

```
SELECT CHECK_NUM, BOOK_NUM, PAT_ID, CHECK_OUT_DATE,  
CHECK_DUE_DATE  
FROM CHECKOUT  
ORDER BY CHECK_OUT_DATE DESC, CHECK_NUM;
```

#### CODE 64

Write a query to display the book title, year, and subject for every book. Sort the results by book subject in ascending order, year in descending order, and then title in ascending order (Partial results are shown in *Figure P7.64*).

```
SELECT BOOK_TITLE, BOOK_YEAR, BOOK_SUBJECT  
FROM BOOK  
ORDER BY BOOK_SUBJECT, BOOK_YEAR DESC, BOOK_TITLE;
```

#### CODE 65

Write a query to display the book number, title, and cost for all books that cost \$59.95 sorted by book number (*Figure P7.65*).

```
SELECT BOOK_NUM, BOOK_TITLE, BOOK_COST  
FROM BOOK  
WHERE BOOK_COST=59.95  
ORDER BY BOOK_NUM;
```

#### CODE 66

Write a query to display the book number, title, and cost for all books in the "Database" subject sorted by book number (*Figure P7.66*).

```
SELECT BOOK_NUM, BOOK_TITLE, BOOK_COST
FROM BOOK
WHERE BOOK_SUBJECT="Database"
ORDER BY BOOK_NUM;
```

#### CODE 67

Write a query to display the checkout number, book number, and checkout date of all books checked out before April 5, 2017 sorted by checkout number (*Figure P7.67*).

```
SELECT CHECK_NUM, BOOK_NUM, CHECK_OUT_DATE
FROM CHECKOUT
WHERE CHECK_OUT_DATE < "2017-04-05"
ORDER BY CHECK_NUM;
```

#### CODE 68

Write a query to display the book number, title, and year of all books published after 2015 and on the "Programming" subject sorted by book number (*Figure P7.68*).

```
SELECT BOOK_NUM, BOOK_TITLE, BOOK_YEAR
FROM BOOK
WHERE BOOK_YEAR > '2015' and BOOK_SUBJECT = 'Programming'
ORDER BY BOOK_NUM;
```

#### Code 69

Write a query to display the book number, title, subject, and cost for all books that are on the subjects of "Middleware" or "Cloud", and that cost more than \$70 sorted by book number (*Figure P7.69*).

```
SELECT BOOK_NUM, BOOK_TITLE, BOOK_SUBJECT, BOOK_COST
FROM BOOK
WHERE (BOOK_SUBJECT = 'Middleware' OR BOOK_SUBJECT = 'Cloud') AND
BOOK_COST > 70
ORDER BY BOOK_NUM;
```

#### CODE 70

Write a query to display the author ID, first name, last name, and year of birth for all authors born in the decade of the 1980s sorted by author

```
SELECT AU_ID, AU_FNAME, AU_LNAME, AU_BIRTHYEAR
FROM AUTHOR
WHERE AU_BIRTHYEAR BETWEEN 1980 AND 1989
ORDER BY AU_ID
```

#### CODE 71

Write a query to display the book number, title, and subject for all books that contain the word “Database” in the title, regardless of how it is capitalized. Sort the results by book number (*Figure P7.71*).

```
SELECT BOOK_NUM, BOOK_TITLE, BOOK_SUBJECT
FROM BOOK
WHERE BOOK_TITLE LIKE '%Database%'
ORDER BY BOOK_NUM;
```

#### CODE 72

Write a query to display the patron ID, first and last name of all patrons who are students, sorted by patron ID (*Figure P7.72*). (44 rows)

```
SELECT PAT_ID, PAT_FNAME, PAT_LNAME
FROM PATRON
WHERE PAT_TYPE = 'Student'
ORDER BY PAT_ID
```

#### CODE 73

Write a query to display the patron ID, first and last name, and patron type for all patrons whose last name begins with the letter “C”, sorted by patron ID (*Figure P7.73*).

```
SELECT PAT_ID, PAT_FNAME, PAT_LNAME, PAT_TYPE
FROM PATRON
WHERE PAT_LNAME LIKE 'C%'
ORDER BY PAT_ID
```

#### CODE 74

Write a query to display the author ID, first and last name of all authors whose year of birth is unknown. Sort the results by author ID (*Figure P7.74*).

```
SELECT AU_ID, AU_FNAME, AU_LNAME
FROM AUTHOR
WHERE AU_BIRTHYEAR IS NULL
ORDER BY AU_ID;
```

#### CODE 75

Write a query to display the author ID, first and last name of all authors whose year of birth is known. Ensure the results are sorted by author ID (*Figure P7.75*).

```
SELECT AU_ID, AU_FNAME, AU_LNAME
FROM AUTHOR
WHERE AU_BIRTHYEAR IS NOT NULL
ORDER BY AU_ID
```

#### CODE 76

Write a query to display the checkout number, book number, patron ID, checkout date, and due date for all checkouts that have not yet been returned. Sort the results by book number (*Figure P7.76*).

```
SELECT CHECK_NUM, BOOK_NUM, PAT_ID, CHECK_OUT_DATE,  
CHECK_DUE_DATE  
FROM CHECKOUT  
WHERE CHECK_IN_DATE IS NULL  
ORDER BY BOOK_NUM
```

#### CODE 77

Write a query to display the author ID, first name, last name, and year of birth for all authors. Sort the results in descending order by year of birth, and then in ascending order by last name (Partial results are shown in *Figure P7.77*).

```
SELECT AU_ID, AU_FNAME, AU_LNAME, AU_BIRTHYEAR  
FROM AUTHOR  
ORDER BY AU_BIRTHYEAR DESC, AU_LNAME
```

#### CODE 78

Write a query to display the number of books in the FACT system (*Figure P7.78*).

```
SELECT Count(BOOK_NUM) AS "Number of Books"  
FROM BOOK;
```

#### CODE 79

Write a query to display the number of different book subjects in the FACT system (*Figure P7.79*).

```
SELECT Count(DISTINCT BOOK_SUBJECT) AS "Number of Subjects"  
FROM BOOK;
```

#### CODE 80

Write a query to display the number of books that are available (not currently checked out) (*Figure P7.80*).

```
SELECT Count(BOOK_NUM) AS "Available Books"  
FROM BOOK  
WHERE PAT_ID IS NULL
```

#### CODE 81

Write a query to display the highest book cost in the system (*Figure P7.81*).

```
SELECT Max(BOOK_COST) AS "Most Expensive"  
FROM BOOK;
```

#### CODE 82

Write a query to display the lowest book cost in the system (*Figure P7.82*).

```
SELECT Min(BOOK_COST) AS "Least Expensive"
FROM BOOK;
```

CODE 83

Write a query to display the number of different patrons who have ever checked out a book (*Figure P7.83*).

```
SELECT Count(DISTINCT PAT_ID) AS "DIFFERENT PATRONS"
FROM CHECKOUT;
```

CODE 84

Write a query to display the subject and the number of books in each subject. Sort the results by the number of books in descending order and then by subject name in ascending order (*Figure P7.84*).

```
SELECT BOOK_SUBJECT, Count(*) AS "Books In Subject"
FROM BOOK
GROUP BY BOOK_SUBJECT
ORDER BY Count(*) DESC, BOOK_SUBJECT;
```

CODE 85

Write a query to display the author ID and the number of books written by that author. Sort the results in descending order by number of books, then in ascending order by author ID (*Figure P7.85*).

```
SELECT AU_ID, COUNT(*) AS 'Books Written'
FROM WRITES
GROUP BY AU_ID
ORDER BY COUNT(*) DESC, AU_ID;
```

Question 86:

Write a query to display the total value of all books in the library (*Figure P7.86*).

```
SELECT SUM(BOOK_COST) AS "Library Value"
FROM BOOK;
```

Question 87:

Write a query to display the patron ID, book number, and days kept for each checkout. "Days Kept" is the difference from the date on which the book is returned to the date it was checked out. Sort the results by days kept in descending order, then by patron ID, and then by book number (*Figure P7.87*). (68 rows)

```
SELECT PAT_ID AS "PATRON", BOOK_NUM AS "BOOK", CHECK_IN_DATE -
CHECK_OUT_DATE AS "Days Kept"
FROM CHECKOUT
ORDER BY CHECK_IN_DATE - CHECK_OUT_DATE DESC, PAT_ID, BOOK_NUM;
```

Question 88:

Write a query to display the patron ID, patron full name, and patron type for each patron

sorted by patron ID (Figure P7.88). (50 rows)

```
SELECT PAT_ID, CONCAT(PAT_FNAME, ' ', PAT_LNAME) AS "Patron Name",  
PAT_TYPE  
FROM PATRON  
ORDER BY PAT_ID;
```

Question 89:

Write a query to display the book number, title with year, and subject for each book.

Sort the results by the book number (Figure P7.89). (20 rows)

```
SELECT BOOK_NUM, CONCAT(BOOK_TITLE, ' (' , BOOK_YEAR, ')') AS "BOOK",  
BOOK_SUBJECT  
FROM BOOK  
ORDER BY BOOK_NUM;
```

CODE 90

Write a query to display the author last name, author first name, and book number for each book written by that author. Sort the results by author last name, first name, and then book number (Figure P7.90). (25 rows)

```
SELECT AU_LNAME, AU_FNAME, BOOK_NUM  
FROM AUTHOR  
JOIN WRITES ON AUTHOR.AU_ID = WRITES.AU_ID  
ORDER BY AU_LNAME, AU_FNAME, BOOK_NUM
```

CODE 91

Write a query to display the author ID, book number, title, and subject for each book. Sort the results by book number and then author ID (Figure P7.91). (25 rows)

```
SELECT AU_ID, BOOK.BOOK_NUM, BOOK_TITLE, BOOK_SUBJECT  
FROM BOOK  
JOIN WRITES ON BOOK.BOOK_NUM = WRITES.BOOK_NUM  
ORDER BY BOOK_NUM, AU_ID;
```

CODE 92

Write a query to display the author last name, first name, book title, and replacement cost for each book. Sort the results by book number and then author ID (Figure P7.92). (25 rows)

```
SELECT AU_LNAME, AU_FNAME, BOOK_TITLE, BOOK_COST  
FROM AUTHOR  
JOIN WRITES ON AUTHOR.AU_ID = WRITES.AU_ID  
JOIN BOOK ON WRITES.BOOK_NUM = BOOK.BOOK_NUM  
ORDER BY BOOK.BOOK_NUM, WRITES.AU_ID
```

#### CODE 93

Write a query to display the patron ID, book number, patron first name and last name, and book title for all currently checked out books. (Remember to use the redundant relationship described in the assignment instructions for current checkouts.) Sort the output by patron last name and book title (*Figure P7.93*).

```
SELECT BOOK.PAT_ID, BOOK_NUM, PAT_FNAME, PAT_LNAME, BOOK_TITLE
FROM PATRON
JOIN BOOK ON PATRON.PAT_ID = BOOK.PAT_ID
ORDER BY PAT_LNAME, BOOK_TITLE;
```

#### CODE 96

Write a query to display the author ID, first and last name, book number, and book title of all books in the subject "Cloud". Sort the results by book title and then by author last name (*Figure P7.96*).

```
SELECT AUTHOR.AU_ID, AU_FNAME, AU_LNAME, BOOK.BOOK_NUM,
BOOK_TITLE
FROM AUTHOR
JOIN WRITES ON AUTHOR.AU_ID = WRITES.AU_ID
JOIN BOOK ON WRITES.BOOK_NUM = BOOK.BOOK_NUM
WHERE BOOK_SUBJECT = "Cloud"
ORDER BY BOOK_TITLE, AU_LNAME;
```

#### CODE 97

Write a query to display the book number, title, author last name, author first name, patron ID, last name, and patron type for all books currently checked out to a patron. Sort the results by book title (*Figure P7.97*).

```
SELECT BOOK.BOOK_NUM, BOOK_TITLE, AU_LNAME, AU_FNAME, BOOK.PAT_ID,
PAT_LNAME, PAT_TYPE
FROM BOOK
JOIN WRITES ON BOOK.BOOK_NUM = WRITES.BOOK_NUM
JOIN AUTHOR ON WRITES.AU_ID = AUTHOR.AU_ID
JOIN PATRON ON BOOK.PAT_ID = PATRON.PAT_ID
ORDER BY BOOK_TITLE
```

#### CODE 100

Write a query to display the author ID, author last name, book title, checkout date, and patron last name for all the books written by authors with the last name "Bruer" that have ever been checked out by patrons with the last name "Miles." Sort the results by check out date (*Figure P7.100*).

```

SELECT WRITES.AU_ID, AU_LNAME, BOOK_TITLE, CHECK_OUT_DATE,
PAT_LNAME
FROM AUTHOR
JOIN WRITES ON AUTHOR.AU_ID = WRITES.AU_ID
JOIN BOOK ON WRITES.BOOK_NUM = BOOK.BOOK_NUM
JOIN CHECKOUT ON BOOK.BOOK_NUM = CHECKOUT.BOOK_NUM
JOIN PATRON ON CHECKOUT.PAT_ID = PATRON.PAT_ID
WHERE AU_LNAME = "Brier" AND PAT_LNAME = "Miles"
ORDER BY CHECK_OUT_DATE

```

#### CODE 99

Write a query to display the book number, title, and number of times each book has been checked out. Limit the results to books that have been checked out more than five times. Sort the results in descending order by the number of times checked out and then by title (*Figure P7.99*).

```

SELECT BOOK.BOOK_NUM, BOOK_TITLE, Count(CHECK_NUM) AS "Times Checked Out"
FROM BOOK
JOIN CHECKOUT ON BOOK.BOOK_NUM = CHECKOUT.BOOK_NUM
GROUP BY BOOK_NUM, BOOK_TITLE
HAVING Count(CHECK_NUM) > 5
ORDER BY Count(CHECK_NUM) DESC, BOOK_TITLE;

```

#### CODE 102

Write a query to display the patron ID, last name, number of times that patron has ever checked out a book, and the number of different books the patron has ever checked out. For example, if a given patron has checked out the same book twice, that would count as two checkouts but only one book. Limit the results to only patrons who have made at least three checkouts. Sort the results in descending order by number of books, then in descending order by number of checkouts, and then in ascending order by patron ID (*Figure P7.102*).

```

SELECT PATRON.PAT_ID, PAT_LNAME, COUNT(CHECK_NUM) AS "NUM CHECKOUTS", COUNT(DISTINCT BOOK_NUM) AS "NUM DIFFERENT BOOKS"
FROM PATRON
JOIN CHECKOUT ON PATRON.PAT_ID = CHECKOUT.PAT_ID
GROUP BY PAT_ID, PAT_LNAME
HAVING COUNT(CHECK_NUM) >= 3
ORDER BY COUNT(CHECK_NUM) DESC, COUNT(DISTINCT BOOK_NUM) DESC, PATRON.PAT_ID;

```

#### CODE 103

Write a query to display the average number of days a book is kept during a checkout (*Figure P7.103*).

```
SELECT ROUND(AVG(CHECK_IN_DATE-CHECK_OUT_DATE),2) AS "Average Days  
Kept"  
FROM CHECKOUT
```

#### CODE 104

Write a query to display the patron ID and the average number of days that patron keeps books during a checkout. Limit the results to only patrons who have at least three checkouts. Sort the results in descending order by the average days the book is kept, and then in ascending order by patron ID (*Figure P7.104*).

```
SELECT PATRON.PAT_ID, ROUND(AVG(CHECK_IN_DATE-CHECK_OUT_DATE),2) AS  
"Average Days Kept"  
FROM CHECKOUT  
JOIN PATRON ON CHECKOUT.PAT_ID = PATRON.PAT_ID  
GROUP BY PAT_ID  
HAVING COUNT(CHECK_NUM)>=3  
ORDER BY ROUND(AVG(CHECK_IN_DATE-CHECK_OUT_DATE),2) DESC,  
PATRON.PAT_ID;
```